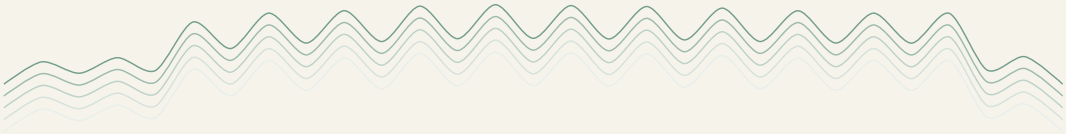
IntelSyst 2026: Intelligent Systems and Smart Technologies

Mitigating hallucinations in low-resource large language models

A case study in Bulgarian

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A 27B model gets this exam question wrong

From a matriculation exam

Разград, Дупница, Троян и София са центрове на:
Razgrad, Dupnitsa, Troyan and Sofia are centres of:

A tobacco B energy C wool textiles D pharmaceuticals

BgGPT-27B with no retrieval

C It answers straight away and gets it wrong.

62.9 %
zero-shot accuracy on 777 exam questions

0.0 %
abstention. It never says “I don’t know”, so its mistakes go unnoticed.

A knowledge graph of Bulgarian geography

What was missing

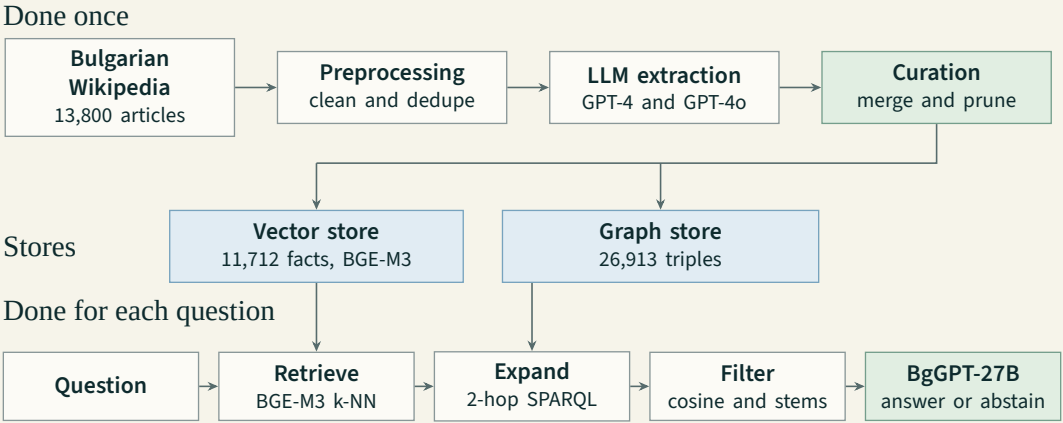
Hallucination is worse in low-resource languages, and elevated rates have been measured for Bulgarian (Berbatova and Salambashev, 2023).

Work on KG-RAG is mostly in English and assumes structured domain resources that Bulgarian does not have.

What we did

1. An LLM turns Bulgarian Wikipedia geography articles into a knowledge graph that fixed rules then clean up.
2. Vector search, a two-hop walk through the graph and a re-ranker for Bulgarian word endings find the facts.
3. We compare the result with zero-shot, Vector RAG and GPT-4o on 777 exam questions.

Offline construction and online retrieval



From 13,800 articles to 11,712 facts

13,800 Wikipedia articles in 21 categories

→ dedupe →

10,700 after cleaning and deduplication

→ extract →

35,600 triples extracted by GPT-4o

→ curate and prune →

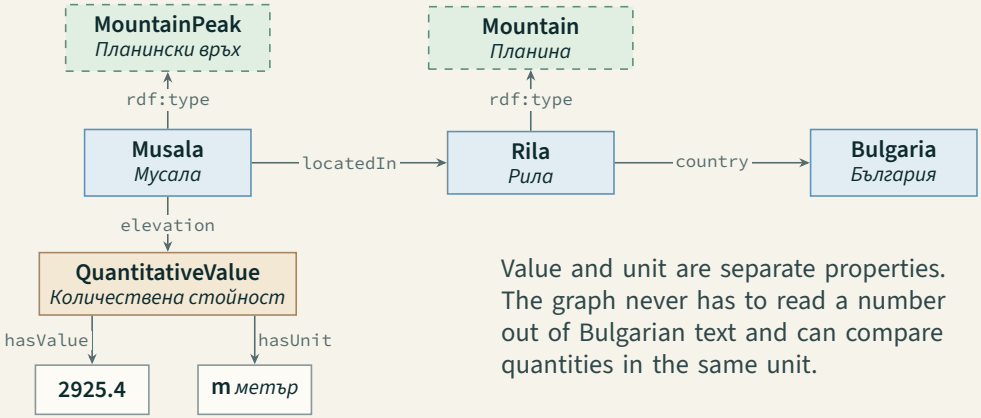
26,913 triples kept for about 6,300 entities

→ verbalise →

11,712 fact strings for the vector store

> The exam questions only decide which entities survive pruning. Every fact comes from Bulgarian Wikipedia, so nothing leaks from the test set. The first three counts are approximate.

A measurement keeps value and unit apart



Value and unit are separate properties. The graph never has to read a number out of Bulgarian text and can compare quantities in the same unit.

Vectors find the start and the graph does the rest

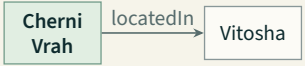
1 Find a starting point

$$S(f,q) = \alpha \cdot \mathrm{sim}(v_f, v_q) + \beta \cdot \mathbb{I}(p_f \in P_{core})$$
$$\alpha = 0.7,\ \beta = 0.3$$

BGE-M3 finds the closest facts. Facts about core relations such as `locatedIn` or `elevation` get a boost.

2 Follow the links

query: “Mountain of Cherni Vrah”



found by vector search

From these entities SPARQL collects everything within two hops. Going further only adds noise.

3 Keep what helps

Cosine similarity narrows the candidates

↓

A keyword match on word stems copes with Bulgarian endings: *планина*, *планината*, *планини*

↓

BgGPT-27B answers from this context or abstains

What we compared and how

SUT-1 Zero-shot	BgGPT-27B on its own
SUT-2 Vector RAG	the five best Wikipedia passages (256-token chunks) after cross-encoder re-ranking
SUT-3 KG-RAG	facts from vector search plus a two-hop walk through the graph
Ref. GPT-4o	a closed frontier model with no grounding that never abstains. An upper bound only

SUT-2 and SUT-3 use the same encoder, generator and strict-grounding prompt. They differ only in *what is retrieved*.

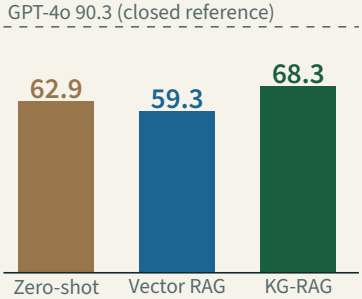


matriculation exams 2008 to 2019 (`reasoning_bg`) — scraped from exams 2019 to 2026

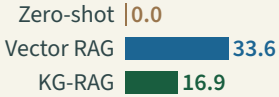
Accuracy counts an abstention as wrong. Abstention rate is how often the system declines. Evidence recall asks whether at least half of the correct option's content words appear in the retrieved context.

KG-RAG is the most accurate local system

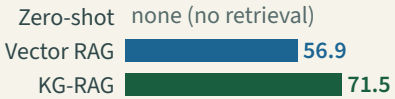
Accuracy (%)



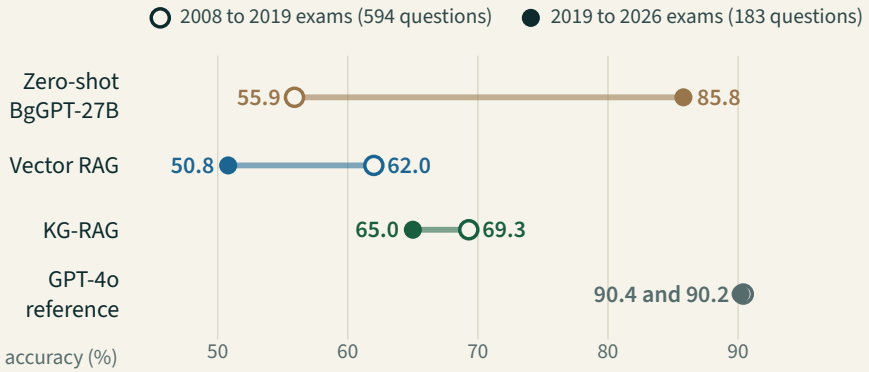
Abstention (lower is better)



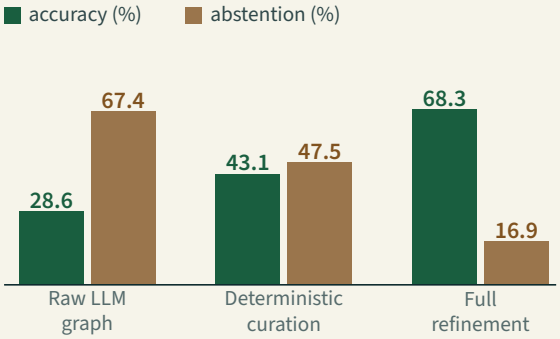
Evidence recall (%)



Zero-shot results depend on the exam era



Curation mattered more than anything else

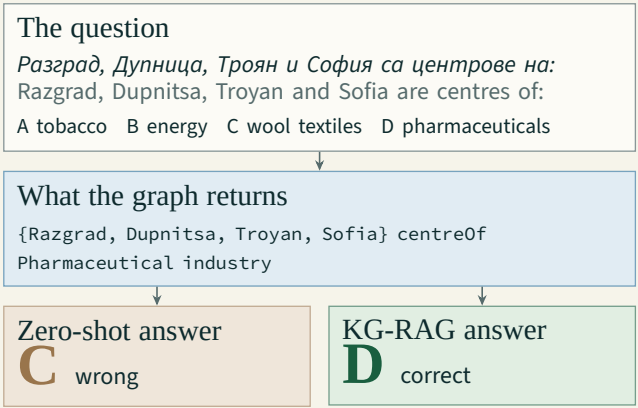


+39.7 points from curation alone, with the same generator and corpus.

An LLM relevance filter turned out to hurt. Accuracy fell from 55.5 to 40.6 and abstention rose from 34.4 to 51.2.

Among re-rankers stem-aware keyword matching reached 62.5, plain keywords 60.0 and a cross-encoder 57.5.

The graph supplies the missing link



How often an answer is right

82.3 %
KG-RAG: answers 83.1 %

89.3 %
Vector RAG: answers 66.4 %

62.9 %
Zero-shot: answers 100 %

Share of correct answers among the questions the system chose to answer.

What we learned

What we found

1. Curation took accuracy from 28.6 to 68.3 percent with the same generator and corpus.
2. Triples surface more of the right evidence than passages do. Recall is 71.5 against 56.9.
3. The model abstains instead of giving an unsupported answer.

What is still missing

- The strict-grounding prompt is over-cautious and sometimes abstains when the evidence is there.
- The graph is static and covers only geography.
- So far we tested one generator model.

Explore the graph and read the code



Graph explorer

`geographbg.vercel.app`

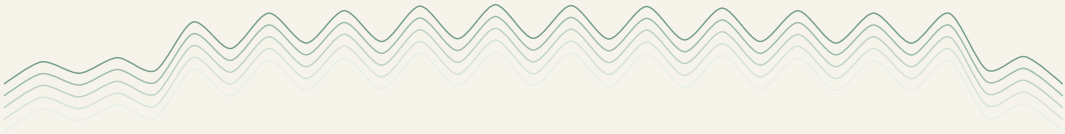


Code and data

`github.com/yonitoo/GeoGraph`

Thank you

Supported by project UNITe BG16RFPR002-1.014-0004, funded by PRIDST.



Full results

	Zero-shot	Vector RAG	KG-RAG	GPT-4o (ref.)
Overall accuracy	62.9	59.3	**68.3**	90.3
Historical accuracy	55.9	62.0	**69.3**	90.4
New accuracy	**85.8**	50.8	65.0	90.2
Abstention rate	0.0	33.6	**16.9**	0.0
Evidence recall	–	56.9	**71.5**	–

Vector RAG chunking	Accuracy	Abstention	Evidence recall
Fixed-size, 256 tokens (33.8k chunks)	**59.3**	**33.6**	**56.9**
Structure-aware (61.4k)	52.6	40.8	49.0
Parent-child (83.3k indexed)	51.1	42.2	45.9

All values in percent. Bold marks the best among the open systems.

Retrieval settings

KG-RAG

- Encoder: BGE-M3, which beat a Bulgarian RoBERTa on evidence recall.
- Re-rank score: $0.7 \cdot \text{sim} + 0.3 \cdot \mathbb{I}(p \in P_{core})$ with $|P_{core}| = 10$.
- Expansion: $n = 2$ hops, beam-bounded at hubs.
- Filter: cosine pre-filter, then a stem-aware keyword re-ranker.
- An o4-mini relevance filter was dropped: it cut accuracy from 55.5 to 40.6.

Vector RAG baseline

- Same encoder, generator and strict-grounding prompt.
- The top 30 by cosine similarity are re-ranked with BGE-reranker-v2-m3.
- The top five passages are kept within a 3,000-token budget.
- No SPARQL expansion, so any gap isolates the structured graph.

How to read the comparison

- GPT-4o is a reference only. It is closed, not local, ungrounded and never abstains. The 2008 to 2019 exams are presumably in its pre-training data.
- Evidence recall is a lexical proxy. A question counts as recalled when at least half of the correct option’s content words (longer than 3 characters) appear in the retrieved context. It is applied identically to passages and triples, so it works as a consistent relative lower bound. Three annotators corroborated it on a sample.
- SUT-2 and SUT-3 abstain with a null token when the context is insufficient. SUT-1 answers from parametric memory alone.
- Each configuration was run five times with consistent results. We report a representative run.